

Project Report

Mobile Application Development
(Using Snack Expo And Supabase)

Library Management System (SmartLIB)

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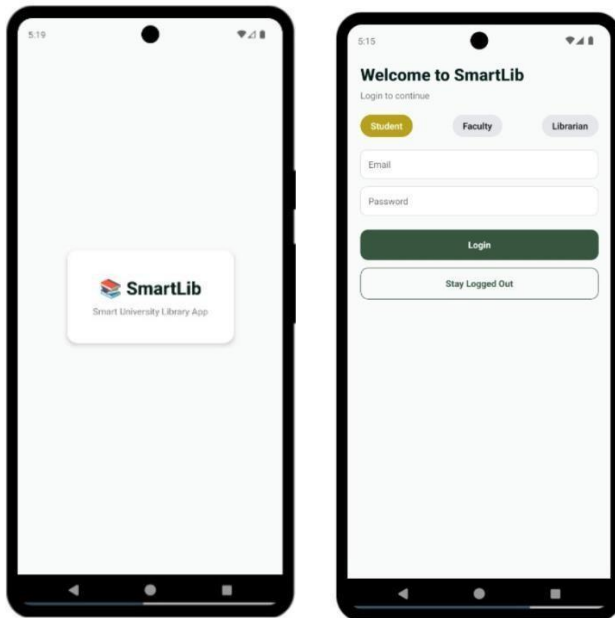
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1. Project Overview

The Library Management System Mobile Application is designed to digitize and streamline library operations. It provides role-based dashboards for librarians, faculty members, and students, enabling each user to perform specific tasks efficiently. The system also allows nonlogged-in users to browse available books in the library.

2. Objectives

- Automate library management processes.
- Provide role-based dashboards for librarian, faculty, and students.
- Enable real-time tracking of book issues, returns, and recommendations.
- Maintain records of member activities and book inventory.
- Ensure secure authentication and role-based access control.
- Use a reliable and scalable backend for database management and real-time updates.



3. Functionalities

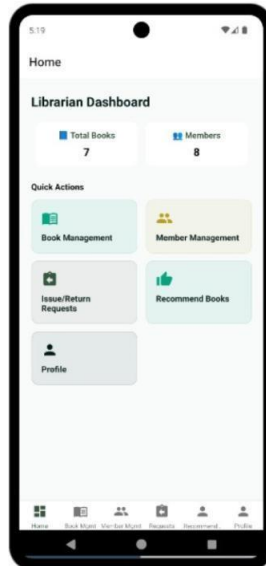
3.1 Librarian Dashboard

Manage Books: Add, update, delete, search, and filter books.

Manage Members: View, add, update, or delete student and faculty members.

Issue Requests Management: Approve or reject book issue requests; track issued books and due dates.

Profile Management: Update personal information and login credentials.



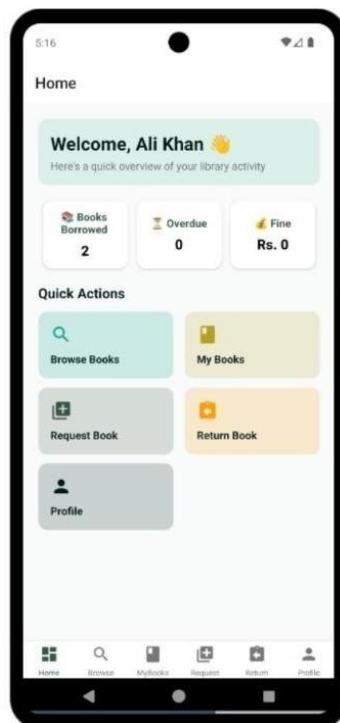
3.2 Student Dashboard

Browse Books: View available books, search by title, author, or category.

Issue Books: Request books for borrowing and track request status.

Return Books: Return borrowed books and view overdue books.

Profile Management: Update personal details.



3.3 Faculty Dashboard

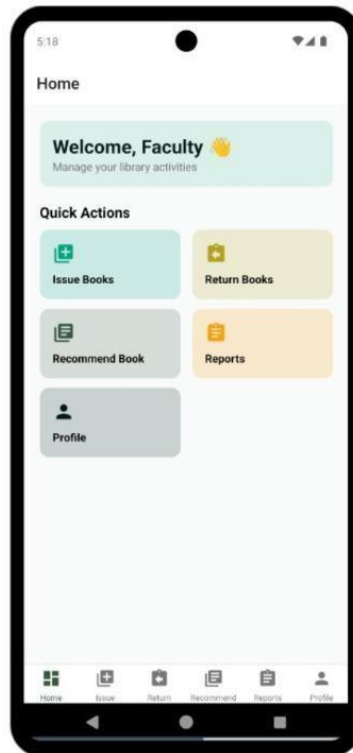
Issue Books: Request books for borrowing.

Return Books: Return previously issued books.

Book Recommendations: Recommend new books to librarians.

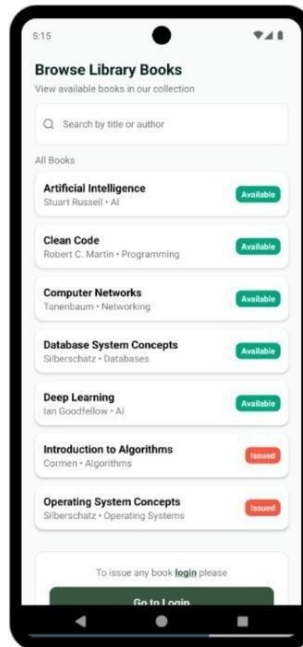
Reports: View issued, returned, and overdue books for their account.

Profile Management: Update personal information.



3.4 Public Access (Logged-Out Users)

- Browse books available in the library without logging in.
- Search and filter the library catalog.



4. Functional Requirements

- Role-based dashboards for librarians, faculty, and students.
- Book management (add, update, delete, browse).
- Member management for students and faculty.
- Book issue and return requests.
- Book recommendation system for faculty.
- Reporting system for faculty and librarian.
- Profile management for all users.
- Real-time data updates.
- Public access to library catalog.

5. Non-Functional Requirements

- **Performance:** Quick response for browsing and searching books.
- **Usability:** Intuitive and user-friendly mobile interface.
- **Security:** Secure authentication and role-based access control.
- **Scalability:** Support for adding more users and books.
- **Reliability:** Real-time updates for book availability.
- **Maintainability:** Modular code for easy future enhancements.

6. Database Specification

6.1 Database Used: Supabase

The project uses Supabase as the backend database, which is a cloud-based PostgreSQL service providing authentication, real-time data, and easy API integration.

6.1.1 Reason for choosing Supabase over alternatives like Firebase

- **Relational Data Handling:** Supabase uses PostgreSQL, which naturally supports relational tables (books, members, requests, issued books, recommendations) and complex queries. Firebase is NoSQL, making relational queries more cumbersome.
- **Real-Time Updates:** Provides real-time subscriptions similar to Firebase but with SQL queries.
- **Authentication:** Offers built-in user authentication with roles for students, faculty, and librarians.
- **Open-Source & SQL-based:** Supabase allows full control over the database structure using standard SQL, unlike Firebase, which is proprietary and NoSQL.
- **Scalability & Performance:** Supabase scales easily while keeping consistent performance for queries and updates.
- **Ease of Integration:** Simple RESTful and client library APIs for mobile applications.

6.1.2 Required Changes for switching to another Supabase database

Change the Supabase Project URL

What it is

This URL points to the Supabase project that hosts your database.

What to change

```
const supabaseUrl = "https://NEW_PROJECT_ID.supabase.co";
```

Where to get it

1. Open **Supabase Dashboard**
2. Select the **new project**
3. Go to **Settings** → **API**
4. Copy **Project URL**

Change the Anon Public Key

What it is

A public client-side API key used for:

- Reading data
- Writing data (based on RLS rules)
- Authentication

What to change

```
const supabaseAnonKey = "NEW_PUBLIC_ANON_KEY";
```

Where to get it

1. Supabase Dashboard
2. Select the **new project**
3. Go to **Settings** → **API**
4. Copy **anon public** key

Important

- Use **anon public**, NOT **service_role**

- service_role must never be used in frontend code

6.2 Tables Used

- **Books** – Stores book information.
- **Issued Books** – Tracks issued books and their status.
- **Book Recommendation** – Faculty-recommended books.
- **Members** – Stores student and faculty information.
- **Requests** – Tracks issue and return requests.

6.3 Tables Relationship:



7. Conclusion

The Library Management System Mobile Application successfully automates library operations while providing role-based access for librarians, faculty, and students. Using Supabase ensures a robust, real-time, and scalable backend with relational data support, secure authentication, and efficient query handling. The application improves usability, transparency, and efficiency in managing library resources, and forms a solid foundation for future enhancements.